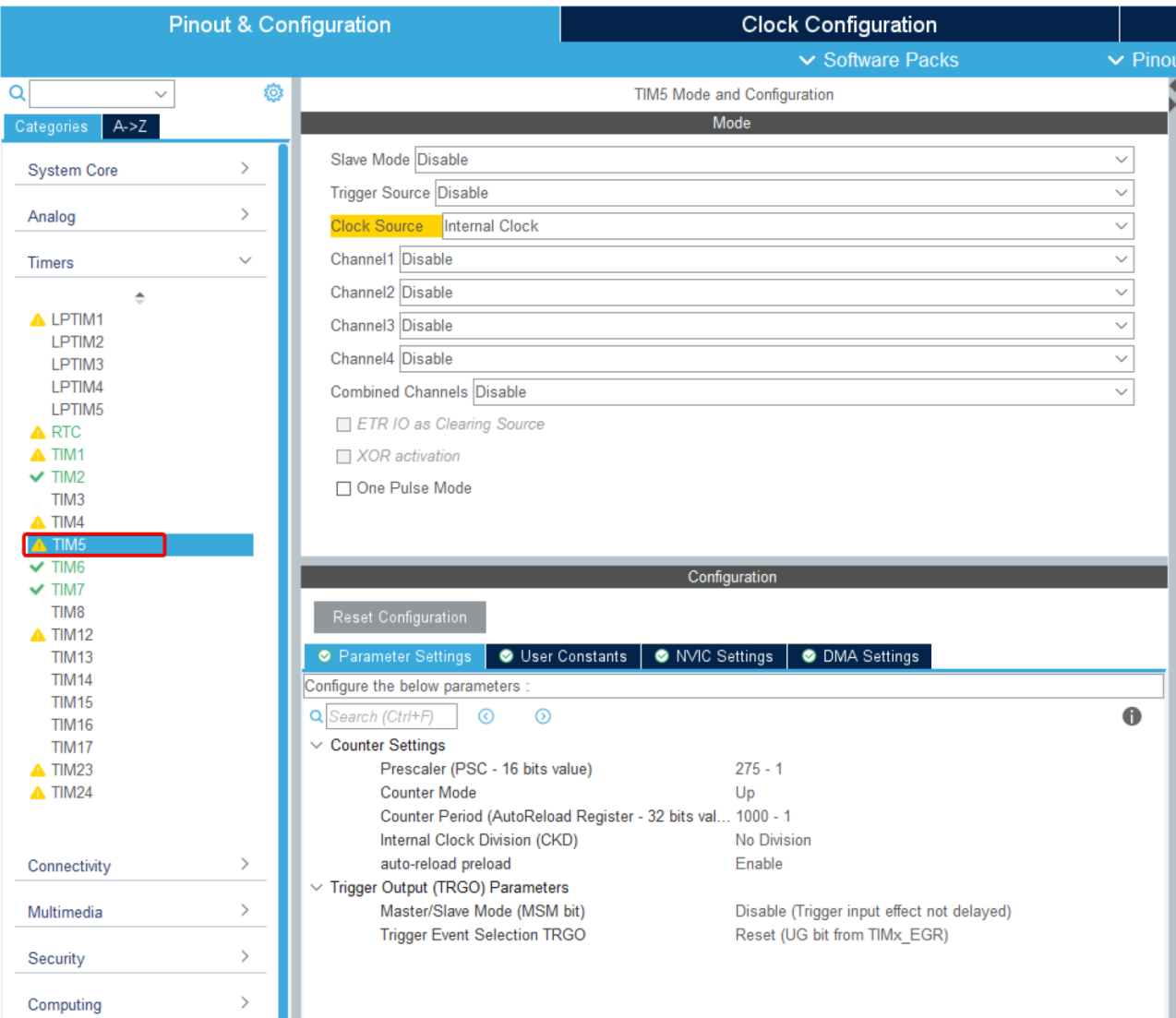


1. 文档说明

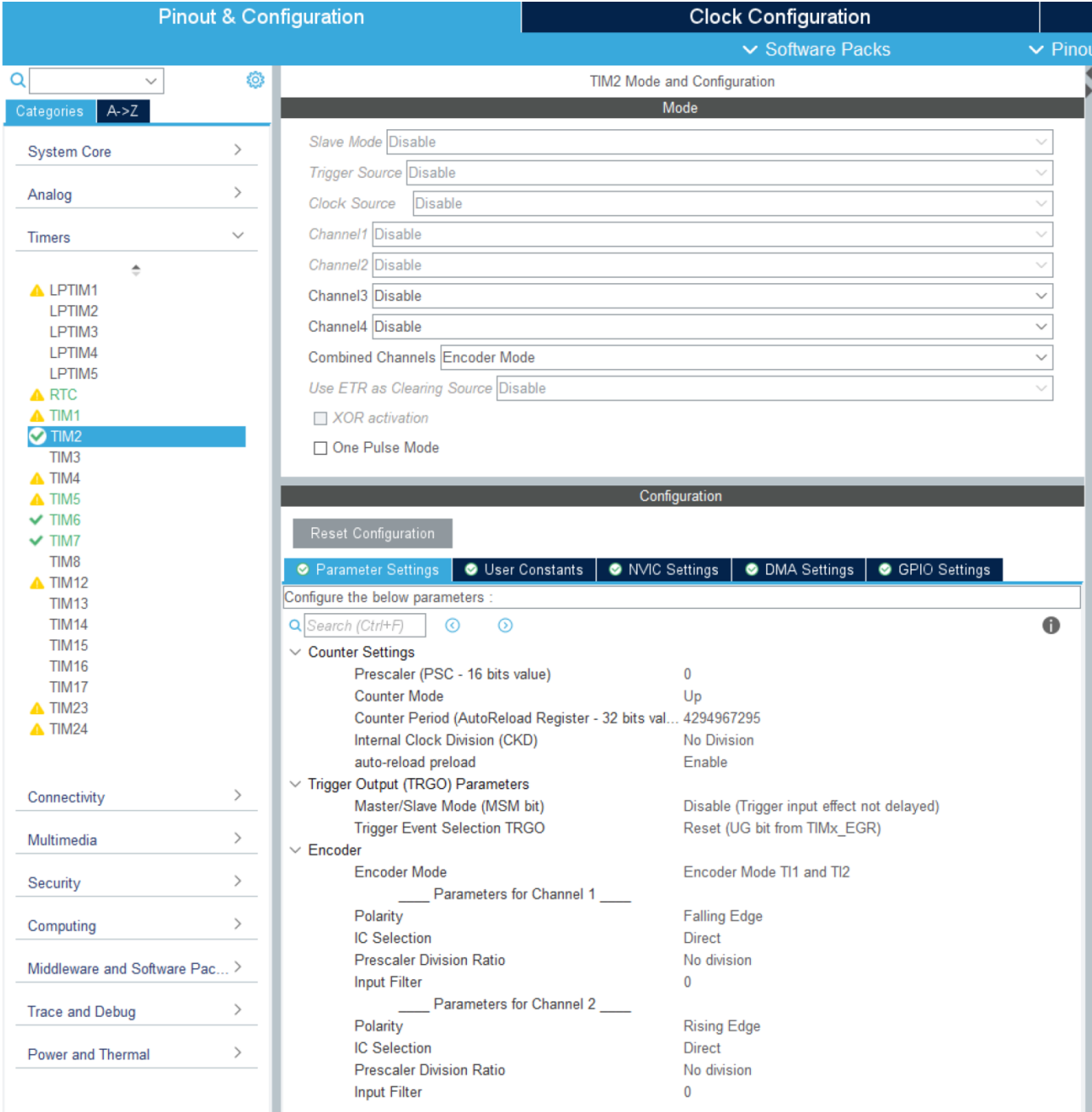
- 本demo针对业务变更情况，此时需要更改片上外设功能配置，并需要编写设备驱动、接口程序
- 本demo拟实现如下功能：
 - 将ethernetcat周期性ECAT_CheckTimer() 功能的时基由TIM2变更为TIM5
 - 将TIM2配置为编码器模式，对外部编码器脉冲进行计数
 - 基础分支：a13a11836aeb6d78a91e6443d3cc124d17173a15
 - demo分支：

2. 修改方式

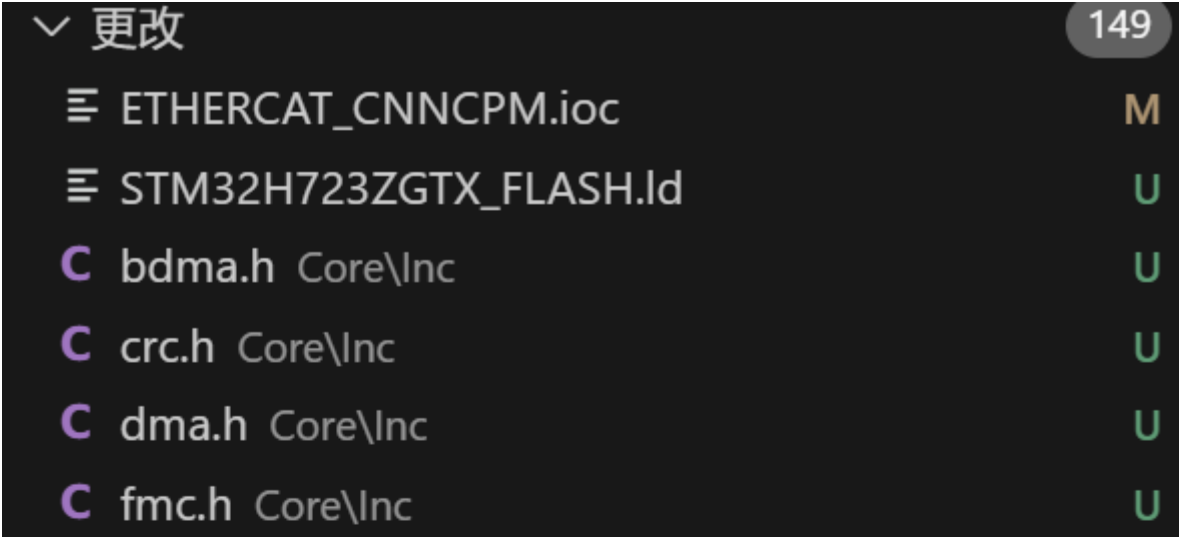
- 配置TIM5，参数与TIM2保持一致



- 修改TIM2为编码器模式，并配置编码器输入引脚为PA0和PA1



- STM32CubeMX重新生成代码，在使用GIT管理工程项目时，可看到新生成的工程相对于原工程有149个文件变更。在工程目录下，可发现新增/更改的文件或文件夹后面有点状或者M/U/D字符样式标识，即表示相关文件或目录下有新增/修改内容



C	FreeRTOSConfig.h	Core\Inc	U
C	gpio.h	Core\Inc	U
C	iwdg.h	Core\Inc	U
C	main.h	Core\Inc	U
C	mdma.h	Core\Inc	U
C	octospi.h	Core\Inc	U
C	rtc.h	Core\Inc	U
C	spi.h	Core\Inc	U
C	stm32h7xx_hal_conf.h	Core\Inc	U
C	stm32h7xx_it.h	Core\Inc	U
C	tim.h	Core\Inc	U
C	usart.h	Core\Inc	U
C	bdma.c	Core\Src	U
C	crc.c	Core\Src	U
C	dma.c	Core\Src	U
C	fmc.c	Core\Src	U
C	freertos.c	Core\Src	U
C	gpio.c	Core\Src	U
C	iwdg.c	Core\Src	U
C	main.c	Core\Src	U
C	mdma.c	Core\Src	U
C	octospi.c	Core\Src	U
C	rtc.c	Core\Src	U
C	spi.c	Core\Src	U
C	stm32h7xx_hal_msp.c	Core\Src	U
C	stm32h7xx_it.c	Core\Src	U

C	system_stm32h7xx.c	Core\Src	U
C	tim.c	Core\Src	U
C	usart.c	Core\Src	U
🔑	LICENSE.txt	drivers\CMSIS	U
🔑	LICENSE.txt	drivers\CMSIS\Device\ST\STM32H7xx	U
C	stm32h7xx.h	drivers\CMSIS\Device\ST\STM32H7x...	U
C	stm32h723xx.h	drivers\CMSIS\Device\ST\STM32...	U
C	system_stm32h7xx.h	drivers\CMSIS\Device\ST\S...	U
C	cmsis_armcc.h	drivers\CMSIS\Include	U
C	cmsis_armclang_ltm.h	drivers\CMSIS\Include	U
C	cmsis_armclang.h	drivers\CMSIS\Include	U
C	cmsis_compiler.h	drivers\CMSIS\Include	U
C	cmsis_gcc.h	drivers\CMSIS\Include	U
C	cmsis_iccarm.h	drivers\CMSIS\Include	U
C	cmsis_version.h	drivers\CMSIS\Include	U
C	core_armv8mbl.h	drivers\CMSIS\Include	U
C	core_armv8mml.h	drivers\CMSIS\Include	U
C	core_armv81mml.h	drivers\CMSIS\Include	U
C	core_cm0.h	drivers\CMSIS\Include	U
C	core_cm0plus.h	drivers\CMSIS\Include	U
C	core_cm1.h	drivers\CMSIS\Include	U
C	core_cm3.h	drivers\CMSIS\Include	U
C	core_cm4.h	drivers\CMSIS\Include	U
C	core_cm7.h	drivers\CMSIS\Include	U
C	core_cm23.h	drivers\CMSIS\Include	U
C	core_cm33.h	drivers\CMSIS\Include	U

```

C core_cm35p.h drivers\CMSIS\Include U
C core_sc000.h drivers\CMSIS\Include U
C core_sc300.h drivers\CMSIS\Include U
C mpu_armv7.h drivers\CMSIS\Include U
C mpu_armv8.h drivers\CMSIS\Include U
C tz_context.h drivers\CMSIS\Include U
🔑 LICENSE.txt drivers\STM32H7xx_HAL_Driver U
C stm32h7xx_hal_cortex.h drivers\STM32H7xx_HA... U
C stm32h7xx_hal_crc_ex.h drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_crc.h drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_hal_def.h drivers\STM32H7xx_HAL_D... U
C stm32h7xx_hal_dma_ex.h drivers\STM32H7xx_H... U
C stm32h7xx_hal_dma.h drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_exti.h drivers\STM32H7xx_HAL_D... U
C stm32h7xx_hal_flash_ex.h drivers\STM32H7xx_H... U
C stm32h7xx_hal_flash.h drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_gpio_ex.h drivers\STM32H7xx_H... U
C stm32h7xx_hal_gpio.h drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_hsem.h drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_i2c_ex.h drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_i2c.h drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_hal_iwdg.h drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_mdma.h drivers\STM32H7xx_HA... U
C stm32h7xx_hal_ospi.h drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_pwr_ex.h drivers\STM32H7xx_HA... U

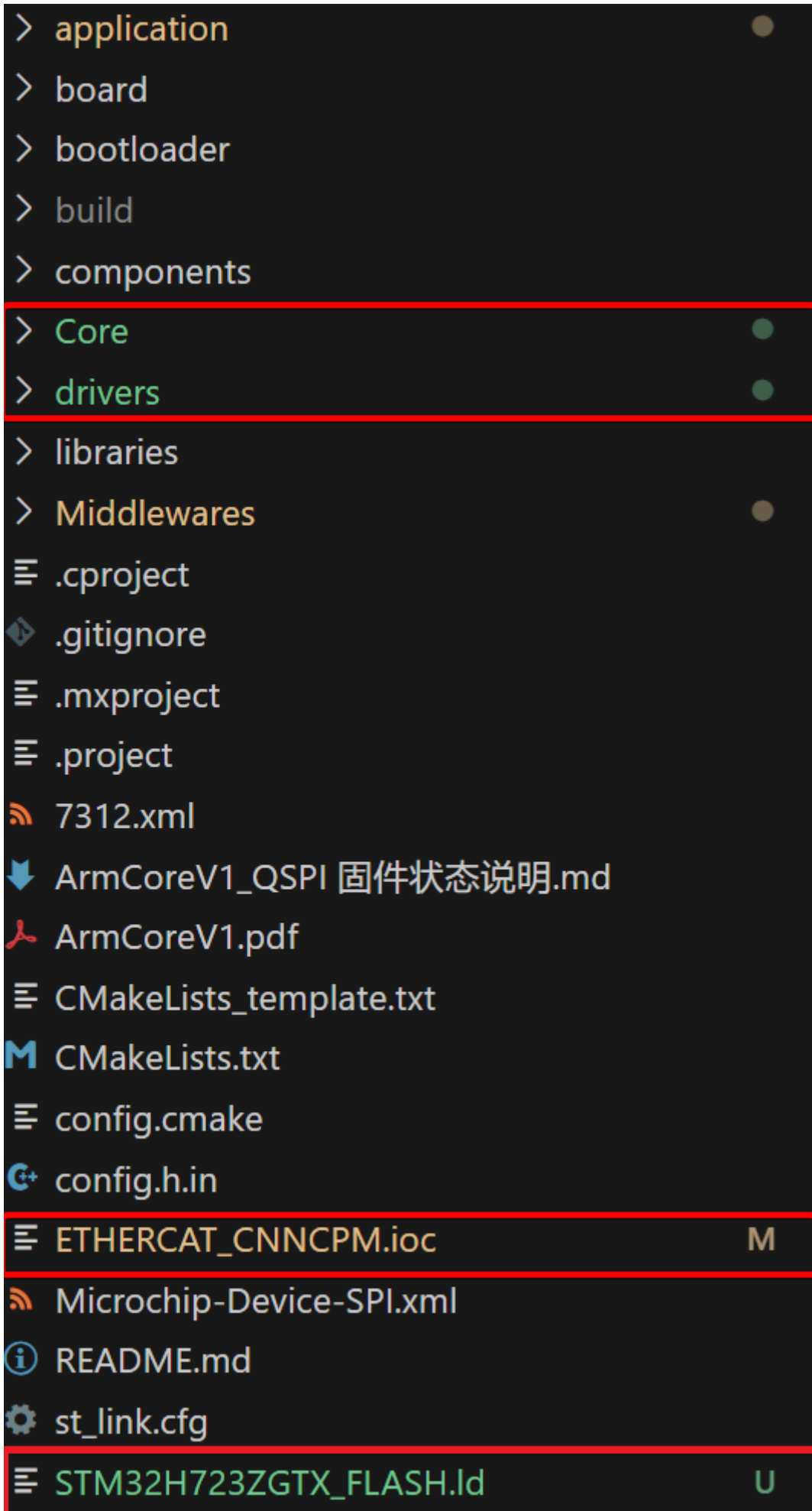
```

```
C stm32h7xx_hal_pwr.h drivers\STM32H7xx_HAL_D... U
C stm32h7xx_hal_rcc_ex.h drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_rcc.h drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_hal_rtc_ex.h drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_rtc.h drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_hal_sdram.h drivers\STM32H7xx_HA... U
C stm32h7xx_hal_spi_ex.h drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_spi.h drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_hal_tim_ex.h drivers\STM32H7xx_HA... U
C stm32h7xx_hal_tim.h drivers\STM32H7xx_HAL_D... U
C stm32h7xx_hal_uart_ex.h drivers\STM32H7xx_H... U
C stm32h7xx_hal_uart.h drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal.h drivers\STM32H7xx_HAL_Driver\... U
C stm32h7xx_ll_bus.h drivers\STM32H7xx_HAL_Dri... U
C stm32h7xx_ll_cortex.h drivers\STM32H7xx_HAL_... U
C stm32h7xx_ll_crc.h drivers\STM32H7xx_HAL_Driv... U
C stm32h7xx_ll_crs.h drivers\STM32H7xx_HAL_Driv... U
C stm32h7xx_ll_dma.h drivers\STM32H7xx_HAL_Dri... U
C stm32h7xx_ll_dmamux.h drivers\STM32H7xx_HA... U
C stm32h7xx_ll_exti.h drivers\STM32H7xx_HAL_Dri... U
C stm32h7xx_ll_fmc.h drivers\STM32H7xx_HAL_Dri... U
C stm32h7xx_ll_gpio.h drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_ll_hsem.h drivers\STM32H7xx_HAL_D... U
C stm32h7xx_ll_iwdg.h drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_ll_lpuart.h drivers\STM32H7xx_HAL_... U
C stm32h7xx_ll_pwr.h drivers\STM32H7xx_HAL_Dri... U
```

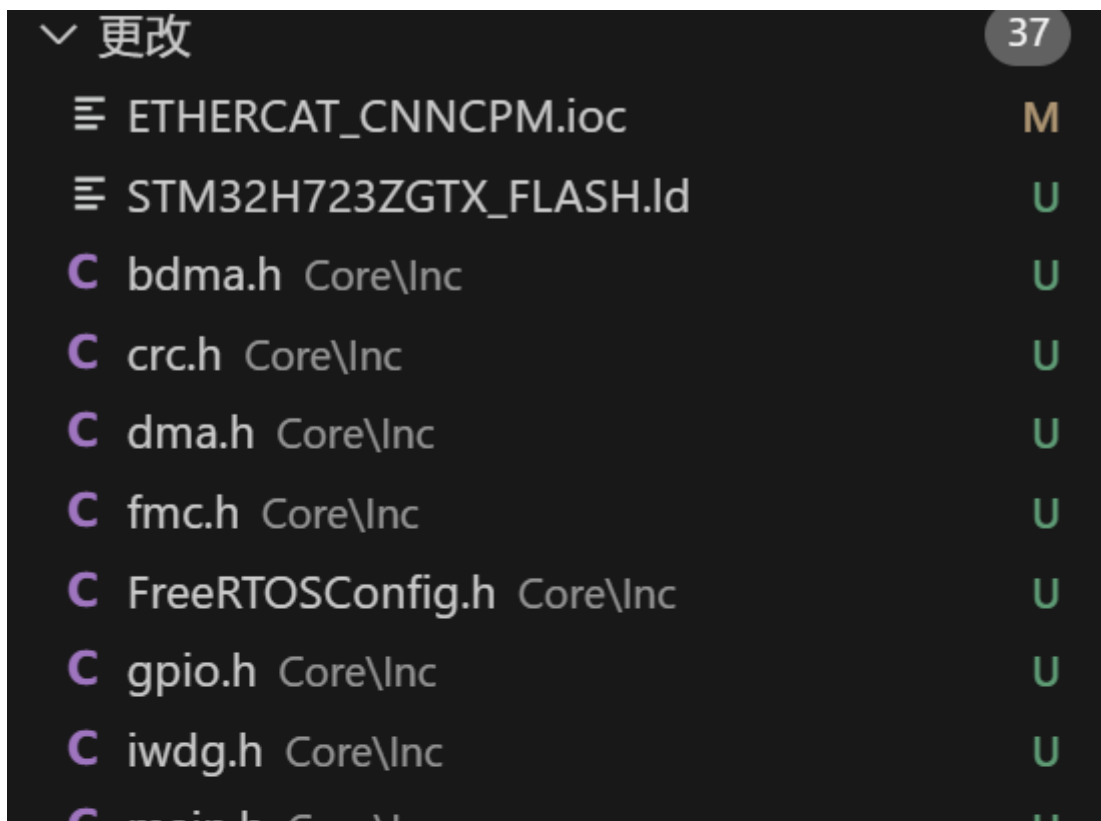
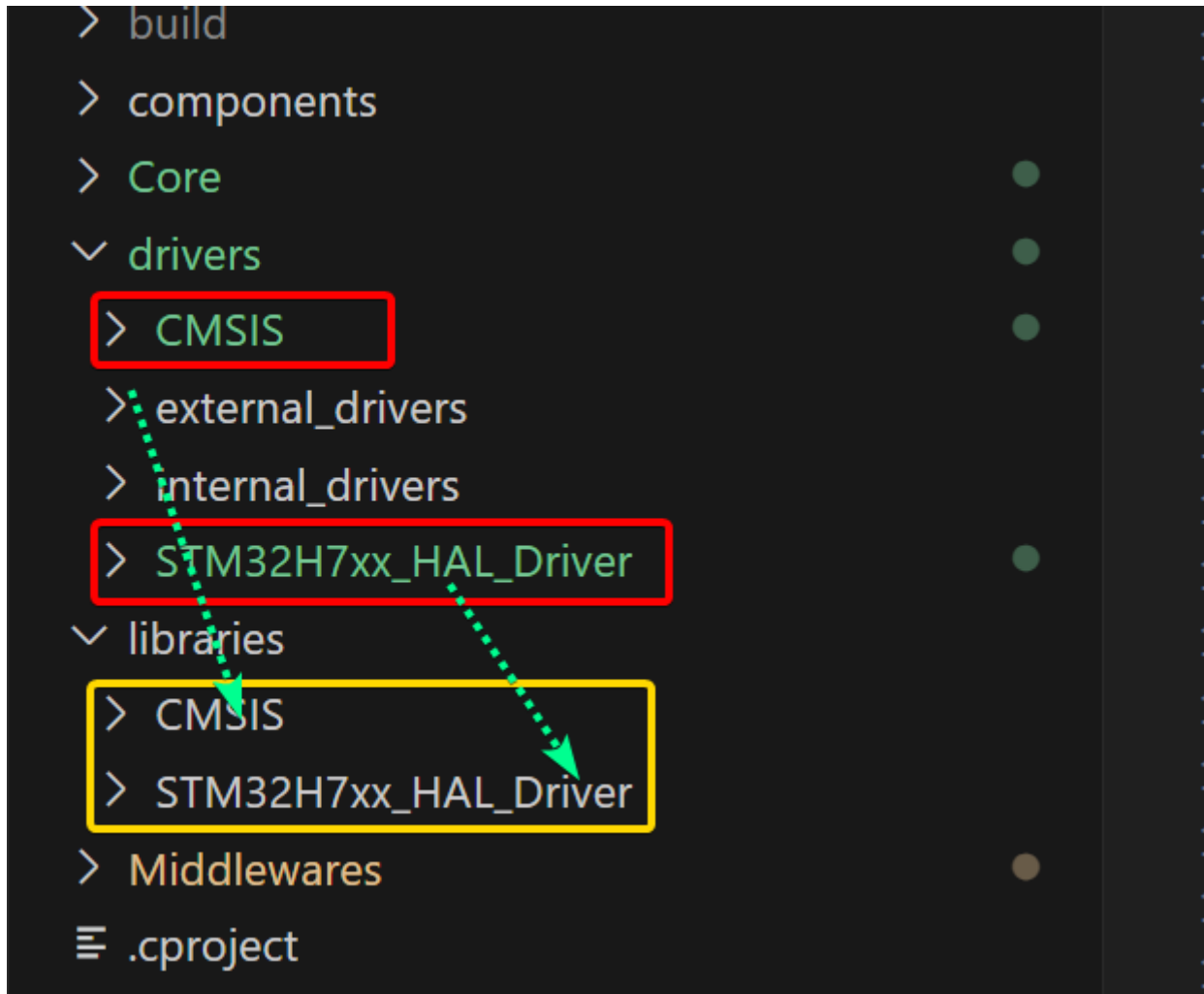


```
C stm32h7xx_ll_rcc.h drivers\STM32H7xx_HAL_Driv... U
C stm32h7xx_ll_rtc.h drivers\STM32H7xx_HAL_Driv... U
C stm32h7xx_ll_spi.h drivers\STM32H7xx_HAL_Driv... U
C stm32h7xx_ll_system.h drivers\STM32H7xx_HAL_... U
C stm32h7xx_ll_tim.h drivers\STM32H7xx_HAL_Driv... U
C stm32h7xx_ll_usart.h drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_ll_utils.h drivers\STM32H7xx_HAL_Dri... U
C stm32_hal_legacy.h drivers\STM32H7xx_HAL_Dri... U
C stm32h7xx_hal_cortex.c drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_crc_ex.c drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_crc.c drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_hal_dma_ex.c drivers\STM32H7xx_H... U
C stm32h7xx_hal_dma.c drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_exti.c drivers\STM32H7xx_HAL_D... U
C stm32h7xx_hal_flash_ex.c drivers\STM32H7xx_H... U
C stm32h7xx_hal_flash.c drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_gpio.c drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_hsem.c drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_i2c_ex.c drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_i2c.c drivers\STM32H7xx_HAL_Dri... U
C stm32h7xx_hal_iwdg.c drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_mdma.c drivers\STM32H7xx_HA... U
C stm32h7xx_hal_ospi.c drivers\STM32H7xx_HAL_... U
C stm32h7xx_hal_pwr_ex.c drivers\STM32H7xx_HA... U
C stm32h7xx_hal_pwr.c drivers\STM32H7xx_HAL_D... U
C stm32h7xx_hal_rcc_ex.c drivers\STM32H7xx_HAL... U
```

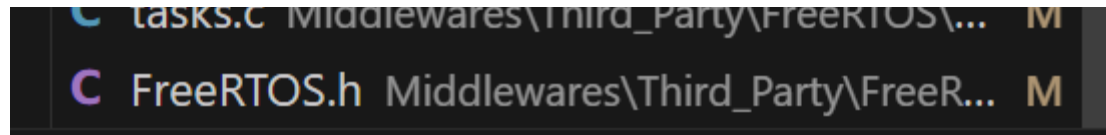
```
C stm32h7xx_hal_rcc.c drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_hal_rtc_ex.c drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_rtc.c drivers\STM32H7xx_HAL_Dri... U
C stm32h7xx_hal_sdram.c drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_spi_ex.c drivers\STM32H7xx_HAL... U
C stm32h7xx_hal_spi.c drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_hal_tim_ex.c drivers\STM32H7xx_HA... U
C stm32h7xx_hal_tim.c drivers\STM32H7xx_HAL_Dr... U
C stm32h7xx_hal_uart_ex.c drivers\STM32H7xx_HA... U
C stm32h7xx_hal_uart.c drivers\STM32H7xx_HAL_D... U
C stm32h7xx_hal.c drivers\STM32H7xx_HAL_Driver\... U
C stm32h7xx_ll_fmc.c drivers\STM32H7xx_HAL_Driv... U
C tasks.c Middlewares\Third_Party\FreeRTOS\Source M
C FreeRTOS.h Middlewares\Third_Party\FreeRTOS\S... M
```

- 将工程drivers目录下CMSIS和STM32H7xx_HAL_Driver文件夹剪切到libraries目录下，直接合并替换libraries目录下原有的文件。经上述操作后，可看到差异文件减少为37个。同时，通过工程目录，新增/更改的文件分布在Core文件夹和Middlewares文件夹下，以及源文件ETHERCAT_CNNCPM.ioc、STM32H723ZGTX_FLASH.ld。



C main.h Core\Inc	U
C mdma.h Core\Inc	U
C octospi.h Core\Inc	U
C rtc.h Core\Inc	U
C spi.h Core\Inc	U
C stm32h7xx_hal_conf.h Core\Inc	U
C stm32h7xx_it.h Core\Inc	U
C tim.h Core\Inc	U
C usart.h Core\Inc	U
C bdma.c Core\Src	U
C crc.c Core\Src	U
C dma.c Core\Src	U
C fmc.c Core\Src	U
C freertos.c Core\Src	U
C gpio.c Core\Src	U
C iwdg.c Core\Src	U
C main.c Core\Src	U
C mdma.c Core\Src	U
C octospi.c Core\Src	U
C rtc.c Core\Src	U
C spi.c Core\Src	U
C stm32h7xx_hal_msp.c Core\Src	U
C stm32h7xx_it.c Core\Src	U
C system_stm32h7xx.c Core\Src	U
C tim.c Core\Src	U
C usart.c Core\Src	U
C tasks.c Middlewares\Third_Party\FreeRTOS	M



> application

> board

> bootloader

> build

> components

> Core

> drivers

> libraries

> Middlewares

≡ .cproject

📁 .gitignore

≡ .mxproject

≡ .project

📶 7312.xml

⬇️ ArmCoreV1_QSPI 固件状态说明.md

📄 ArmCoreV1.pdf

≡ CMakeLists_template.txt

M CMakeLists.txt

≡ config.cmake

⚙️ config.h.in

≡ ETHERCAT_CNNCPM.ioc

M

📶 Microchip-Device-SPI.xml

📄 README.md

⚙️ st_link.cfg

≡ STM32H723ZGTX_FLASH.ld

U

ETHERCAT_CNNCPM.ioc文件是STM32CubeMX的配置文件，记录了硬件的相关配置，**必须保留**，且每次更新硬件配置时均需要更新工程下该文件。

▼ 更改

☰ ETHERCAT_CNNCPM.ioc

☰ STM32H723ZGTx_FLASH.ld

C main.h Core\Inc

C stm32h7xx_it.h Core\Inc

C tim.h Core\Inc

C main.c Core\Src

C stm32h7xx_it.c Core\Src

C tim.c Core\Src

C tasks.c Middlewares\Third_Party\FreeRTOS\Source

C FreeRTOS.h Middlewares\Third_Party\FreeRTOS\S...

10

M

U

U

U

U

U

U

M

M

STM32CubeMX配置文件
需要保留

STM32H723ZGTx_FLASH.ld文件是STM32CubeMX的链接脚本文件，需要将其与board/linker_scripts目录下的同名文件进行对比，并根据实际情况进行修改。因本demo不涉及链接脚本的修改，故直接删除该文件。

更改		10
≡	ETHERCAT_CNNCPM.ioc	M
≡	STM32H723ZGTX_FLASH.ld	U
C	main.h Core\Inc	U
C	stm32h7xx_it.h Core\Inc	U
C	tim.h Core\Inc	U
C	main.c Core\Src	U
C	stm32h7xx_it.c Core\Src	U
C	tim.c Core\Src	U
C	tasks.c Middlewares\Third_Party\FreeRTOS\Source	M
C	FreeRTOS.h Middlewares\Third_Party\FreeRTOS\S...	M

链接文件
删除

Middlewares是STM32CubeMX的Middleware and Software Packs分类菜单生成的中间件文件夹，当前仅放置了FreeRTOS相关文件。由于工程中移植了cm_backtrace组件用于硬件fault追踪。故，需要保留原文件（在相应文件上点击放弃更改，可直接删除更新的内容）。

更改		9
≡	ETHERCAT_CNNCPM.ioc	M
C	main.h Core\Inc	U
C	stm32h7xx_it.h Core\Inc	U
C	tim.h Core\Inc	U
C	main.c Core\Src	U
C	stm32h7xx_it.c Core\Src	U
C	tim.c Core\Src	U
C	tasks.c Middlewares\Third_Party\FreeRTOS\Source	M
C	FreeRTOS.h Middlewares\Third_Party\...	M

放弃更改

将Core目录下, Inc和Src文件夹中的文件, 与drivers/internal_drivers文件夹下同名文件进行对比, 并修改drivers/internal_drivers文件夹下的对应文件内容。本demo只涉及timer的变更, 故只需对比修改tim.h、tim.c、main.h、main.c、stm32h7xx_it.h和stm32h7xx_it.c这几个文件。将Core文件夹下未涉及文件删除后, 如下:

更改		7
ETHERCAT_CNNCPM.ioc		M
main.h	Core\Inc	U
stm32h7xx_it.h	Core\Inc	U
tim.h	Core\Inc	U
main.c	Core\Src	U
stm32h7xx_it.c	Core\Src	U
tim.c	Core\Src	U

- 修改完成后的文件更改情况

更改		6
ETHERCAT_CNNCPM.ioc		M
main.c	application\user_app\common	M
stm32h7xx_it.h	board\board_config\inc	M
9252_HW.c	drivers\external_drivers\lan9252	5, M
tim.c	drivers\internal_drivers\timer	M
tim.h	drivers\internal_drivers\timer	M

- 若CmakeLists.txt文件有修改，则需要重新配置工程（重新生成makefile文件）

libraries

> Middlewares

≡ .cproject

🔒 .gitignore

≡ .mxproject

≡ .project

🔌 7312.xml

📄 ArmCoreV1_QSPI 固件状态说明

📄 ArmCoreV1.pdf

≡ CMakeLists_template.txt

M CMakeLists.txt

≡ config.cmake

🔌 config.h.in

≡ ETHERCAT_CNNCPM.ioc

🔌 Microchip-Device-SPI.xml

📘 README.md

⚙️ st_link.cfg

在侧边打开Ctrl+Enter

打开方式...

在文件资源管理器中显示Shift+Alt+R

在集成终端中打开

🤖 启动ChatGPT中文版

选择以进行比较

Open Changes>

Open on Remote (Web)>

🕒 打开时间线>

剪切Ctrl+X

复制Ctrl+C

复制路径Shift+Alt+C

复制相对路径Ctrl+K Ctrl+Shift+C

Copy Remote File URL

Copy Remote File URL From...

运行测试

调试测试

重命名...F2

删除Delete

Git: View File HistoryAlt+H

生成所有项目

清理所有项目

清理所有项目的重新生成

清理所有项目的重新配置

使用 CMake 调试器清理重新配置所有项目

配置所有项目

右键->重新配置

- 编译工程

```

问题 13  输出  调试控制台  终端  端口  MEMORY  XRTOS  GITLENS

[ 91%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_sdram.c.obj
[ 92%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_spi.c.obj
[ 93%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_spi_ex.c.obj
[ 94%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_tim.c.obj
[ 95%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_tim_ex.c.obj
[ 96%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_uart.c.obj
[ 97%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_hal_uart_ex.c.obj
[ 98%] Building C object CMakeFiles/ETHERCAT_CNNCPM.elf.dir/libraries/STM32H7xx_HAL_Driver/Src/stm32h7xx_ll_fsmc.c.obj
[100%] Linking C executable ETHERCAT_CNNCPM.elf
Memory region      Used Size  Region Size  %age Used
ITCMRAM:           0 GB       64 KB         0.00%
DTCMRAM:           0 GB      128 KB         0.00%
FLASH:            163248 B        1 MB        15.57%
RAM_D1:            129624 B       320 KB        39.56%
RAM_D2:             1 KB        32 KB         3.13%
RAM_D3:            176 B        16 KB         1.07%
RAM_BKP:             6 B         4 KB         0.15%
Building F:/WORK/arm_core_develop/arm_core_restruction/FW-ArmCore/ArmCoreV1_QSPI/build/ETHERCAT_CNNCPM.hex
Building F:/WORK/arm_core_develop/arm_core_restruction/FW-ArmCore/ArmCoreV1_QSPI/build/ETHERCAT_CNNCPM.bin
[100%] Built target ETHERCAT_CNNCPM.elf
PS F:\WORK\arm_core_develop\arm_core_restruction\FW-ArmCore\ArmCoreV1_QSPI\build>

```

- 下载程序测试

```

问题 19  输出  调试控制台  终端  端口  MEMORY  XRTOS  GITLENS

Info : STLINK V2J40S7 (API v2) VID:PID 0483:3748
PS F:\WORK\arm_core_develop\arm_core_restruction\FW-ArmCore\ArmCoreV1_QSPI\build> make flash
Open On-Chip Debugger 0.12.0 (2023-10-02) [https://github.com/sysprogs/openocd]
Licensed under GNU GPL v2
libusb1 09e75e98b4d9ea7909e8837b7a3f00dda4589dc3
For bug reports, read
    http://openocd.org/doc/doxygen/bugs.html
Info : auto-selecting first available session transport "hla_swd". To override use 'transport select <transport>'.
Info : The selected transport took over low-level target control. The results might differ compared to plain JTAG/SWD
Info : clock speed 1800 kHz
Info : STLINK V2J40S7 (API v2) VID:PID 0483:3748
Info : Target voltage: 3.207737
Info : [stm32h7x.cpu0] Cortex-M7 r1p2 processor detected
Info : [stm32h7x.cpu0] target has 8 breakpoints, 4 watchpoints
Info : starting gdb server for stm32h7x.cpu0 on 3333
Info : Listening on port 3333 for gdb connections
[stm32h7x.cpu0] halted due to debug-request, current mode: Thread
xPSR: 0x01000000 pc: 0x08006b48 msp: 0x24050000
** Programming Started **
Info : Device: STM32H72x/73x
Info : flash size probed value 1024k
Info : STM32H7 flash has a single bank
Info : Bank (0) size is 1024 kb, base address is 0x08000000
Info : Padding image section 1 at 0x08027db0 with 16 bytes (bank write end alignment)
Warn : Adding extra erase range, 0x08027dc0 .. 0x0803ffff
Warn : no flash bank found for address 0x38000000
Warn : no flash bank found for address 0x38000000
** Programming Finished **
** Resetting Target **
shutdown command invoked
PS F:\WORK\arm_core_develop\arm_core_restruction\FW-ArmCore\ArmCoreV1_QSPI\build>

```

- 至此，完成硬件配置的修改，下一步根据业务需求，编写相应的设备驱动接口。本demo编码器接口实现可参考SHA-1: b5edeeab0f16bb55d3f87af913d04df3caab39ee下encoder_port.h和encoder_port.c文件